

ActInf Textbook Group ~ Cohort 1 ~ Meeting 18

(Chapter 7, part 2)

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hello cohort one it's meeting 18 we're in our second discussion on chapter seven let's go to the questions and see what we can explore today or see where else to go and or look towards chapter eight but there's any number of ways we can do it so first just on chapter seven does anyone want to uh turn to any of these questions add another question add another reflection or a thought that arose in the last week one question that I um had was are there any ways or sweets for for example taking in an analytical expression and then providing equivalent phrasings that might have other advantages Ali you mean by translating it into natural language or like like what we saw in equation 2.5 and 2.6 like to take in an expression and then output [Music] um isomorphic or I guess not isomorphic but equivalent expressions Expressions that have the exact same value as calculated because some some of the and when we look through the derivation sometimes it's possible to trace the trail but to know which representations of perhaps even the same functional or same term it seems quite relevant actually there's a an AI assistant for deriving formal proofs for mathematical theorems but uh I haven't used it myself but I'm not sure if that's what you mean I'll look at the book the name of that and what just one second I'll post hmm maybe there's a way to reformulate these um specific terms from like a Bayesian mechanics uh formalism like posterior predictive predicted entropy that looks like something that Dalton probably described as well the question is whether you can just interchange it yeah does working with the particular partition enable equations to be operated with more composably because we can know that there are certain operations that are um like always sometimes never going to be valid I'm just kind of asking I don't even know if that's the threat with a habit so last time we talked primarily about the Mouse in the maze and we're going through uh the way that the this the chapter is layering on features of the model so first we saw the mouse just go for it and now we're going to be uh it gets where it gets the queue and this comes to our earlier points about the resolution of the explore exploit trade-off foreign anyone can give a thought while I'm adding it what are

posterior predictive entropy and expected ambiguity the part of the decomposition of epistemic value let me just make sure to say the posterior predictive entropy is the expected surprise or the entropy of the distribution of observations conditioned on a policy so we're in expected free energy world we're talking about evaluating policies with respect to now putting aside pragmatic value we're talking about decompositions of how the informational or the epistemic value of a policy is evaluated so there's two terms here first how dispersed is your distribution of outcomes for that policy one can imagine that all things being equal you would want to select policies that have a tighter distribution of outcomes here we have an expectation interestingly um this is italics e but it's not a fancy e do people think there's a difference that matters or do they think that's a slight error I think there should be a typo yeah I'm going to add it to errata e is italics should be fancy e okay it's the expectation over our hidden State estimates condition on a policy and it's the expectation of the expected ambiguity so entropy of the a matrix functionally how outcomes depend on States so this is saying I want to be I want to be more is it the case that it says I want to be more certain about observations and how they uh map to policy and I want to have a tighter a matrix Eric so um knife question why is there an expectation on the right and now on the left I was also going to ask this isn't entropy already the expectation of surprise just just so another another point of um clarification for myself that I want to make sure this is right is um the um the O tilde is observations and what we're looking at here is a Q so that's something that's that's a distribution observations that will get over policy and that's what's being adjusted in the free energy so we're going to explore the uh the distributions Q and the o tilde's are going to be past observation so we're not not looking forward there we're looking past because those are actual observations they're not predicted through um the the hidden States s so straighten that out are we sure that oh our only past observations I believe that past observations are accounted for with respect to how they influence the prior but does o tilde also include future observations well if so then it has to be an expectation over something that mustn't it like but this is conditioned upon a given policy then these are the this is the time series of observations expected under that policy is the left term the Q expected distribution of observations conditioned on policy Jacob I was just going to uh say that I think I've seen uh this in other parts of the book as well and I think they use the H and then just the expectation notation for entropy kind of interchangeably um okay if they do but then isn't this like a double expectation this is the this is the expectation of surprise on the queue but this is the expectation of the expectation of surprise yeah uh that I don't know okay okay but the left term just um yeah Ali please uh well shouldn't that first line uh I mean

the H the both H is shouldn't both of them be as the argument of the expectation according to equation 7.4 because in equation 7.4 we have the negative epistemic value but it's somehow different from this equation yeah that's it so in 7.4 we see a re statement expected free energy equation two six let's leave pragmatic value to the side for now we see fancy $e q s \tilde{\pi}$ so this is looking like this probably could or should we also see $h p o s h p o s \text{ minus } H Q o \pi$ okay so let's take the negative of the first line the negative of the first line you can kind of just flip the terms it's the expected ambiguity now minus the the posterior predictive entropy an expectation is being taken over that oh sorry sorry my bad I felt the the square brackets I mean it was encompassing the whole expectation term sorry sorry no no you're you're right actually um oh well there's a lot of layers of it I I always write as it's written okay okay yeah yeah okay wait I always color my brackets when I'm trying to do this um okay the the outermost bracket there actually may be a bracket I don't even know let's try to figure it out the outer bracket starts here on the left and it and then it closes okay pragmatic value if we take that as a standalone if we take that as standalone is anyone else seeing a bracket sort of scenario if we look at the brackets as a whole this first bracket closes this right Mouse uh I don't think so the first bracket closes at the end of the $S \tilde{\pi}$ yeah okay okay yeah parenthesis okay okay bracket yeah that's okay well I was confused too I got I I agree that was I was thinking should we evaluate the brackets locally but yes they they are okay good okay so it's exactly the negative of equation seven point eight left side here is the negative of so the first term um let's just say it in terms of picking good policies we want policies are better with more clarity about observation outcomes that's the first term and a tighter a matrix that's the second term middle line this is a KL Divergence between the product two two products of distributions both of them are dealing with o in the first and s in the second the Q distribution in the second part is the same for both so maybe there's some interactions in how distributions of different kinds multiply but it may be fair to interpret this as um the POS is the a matrix and the $q o \pi$ is how observations are shaped by policy or another way to say it might be POS is outcome expectations conditioned upon hidden states of the world whereas this is outcome distributions conditioned on what we do I mean to me that just looks like like base you know base law there because you've got um at least in the in the left side a little bit because you've got condition on S and then s conditioned on π so you're just basically saying you're trying to figure out what o is from π through this intermediary s Delta oh one so another angle would be where's this zero well this is zero when these two are identical the epistemic value of a policy is low when it when the resulting o distribution is being unconditioned by the policy that we choose

whereas for policies that change how we think about the a matrix those policies have higher informational value because POS is kind of like a policy agnostic mapping but we're like drilling down to kind of sub variance partitions of informational value these are different ways to look at what makes a policy have epistemic value the first one line was coming down to uh clarity around the tightness of outcome the second one is related to how policy deviates our understanding of the a matrix now let's turn to okay here's another um italic e not a fancy e thank you Ali for this link I've I've heard of this one the the proof assistant but I've never seen it used or anything yeah this looks pretty pretty interesting I wonder if anyone has done that kind of formal work on proving certain parts of fep and then a more distal question would be if we have formal ontology representations for example with sumo or otherwise could we make proofs around the ontology and the Sumo um suggested upper merged ontology we explored some of those things but didn't didn't go super super far it was just the the it's it's it's yeah take somebody who knows that area the bottom term the bottom line an expectation over q but now our Q is on outcomes not focusing on the hidden States then it's an expectation about outcomes a kale Divergence it's about this is about hidden States this kale Divergence is zero when outcomes are not when in addition to policy outcomes don't change how we think about hidden so when outcomes influence how we think about hidden States this is going to be higher and what what does on the right this is a triangle meaning defined as so q s π o is defined as and this looks like a Bayes theorem situation like kind of that chaining that Eric just mentioned like you have P of O condition on S and then you have S conditioned on π so then those kind of like it's like a way to get from o through S to π whereas this one on the bottom is just like going directly about oh conditioned on π which interestingly is I don't know if this is reflected one of these or or in some subsets of them um one can think about that kind of structural model like the minimum hidden Markov model not even in its full framing but just like what are the three pillars of the Hidden Markov model observations hidden States policy yes Transitions and so on and so what is gained by going through an intermediary hidden state there's going to be situations where the observable contains a hundred percent of the information for policy or we can imagine situations where the ability to choose policies based upon s which is where the actual pomdp is intervening is better whereas in a fully observable Markov decision process you don't have S you're just making decisions on o um kind of just revisiting the point I raised earlier about what $otilda$ is it seems to me that this only works when you can when you have the observations oh tilde I I don't see how that could how it could be anything like looking toward the future because if you actually want to carry out that

um that expect that that you want to work out what Q is or use it you have to compute it and you have to have the observations to compute it and and you know the S tilde you have to have you know actually pipe through use them so help me help me understand if I'm wrong with that okay here would be um one thought the a matrix captures how hidden States map to observations so whether we have a really tight a matrix or whether we have a super dispersed a matrix using the gaussian form for the a matrix ensures that we can always say what the most likely observation is for a given hidden state if we had a bimodal a matrix there are some issues but if we use a gaussian form for the um like which is the laplacian approximation it's more tractable it's monotonic it's more optimizable so for a given s sequences of s we can always derive the expected observations again those might have a lot of uncertainty around them but we can always make a sequence of O 's corresponding to any sequence of s but this this equation makes no expectation at all nothing says nothing about what the form of the distribution is that you use I mean whether it's parametric or not parametric or whatever it's just saying you have a distribution so you know then this is how we evaluate yes is this true for all families of Ali what do you think about that or uh actually it's interesting that uh in the step-by-step active inference paper this epistemic value is defined uh somewhat differently uh I'm not sure if they're exactly equivalent with each other or there's some minor uh differences between them but uh they've just defined the epistemic value only in terms of the expectation over I mean the surprises of I mean the difference between the qnp surprises so we don't have this kind of expression as conditioned on the S tilde here so yeah that's I don't know if they're the same or not I don't know how to copy the image into this chat but the chat of The Gather but or just put it put it in here put it in the um okay yeah um well one one thought is policy is not an open-ended policy in its specification is always going to be of a finite Horizon so if we're considering policies of length 2. um well so so interestingly this is a policy that includes leading up to the present if this is the present I think we had one meme that was like what if my T minus one was was t or something like that but um this is a policy that right is influencing and this is just the variant of four three okay but let's just say that we're considering policy of true future looking length 2. we're exploring different ways in which s is going to unfold with respect to basically B_1 and B_2 not using that in the superscript way to mean like two two parallel bees but just like there's our current there's our estimate now and then there's the way that that we have you know let's just say four affordances so there's four B ones you have now there's going to be four options for S and then each of those four we could apply the second affordance again so there's 16 policies to evaluate each one is defined by

either taking like B1 B3 B1 B1 B4 B4 and for each of those s in the next two time steps we can emit Ollie and then Rohan sorry reminder my hand just stay away I'm sorry rod my hello yes greetings yeah so uh going back to what the posterior distribution is right so how far so is it just that isn't it just that we assume there's a posterior distribution of states like an ideal posterior distribution that we want to that free energy that the free energy principle moves moves the current empirical distribution towards isn't that I thought that's what we were going from the context in chapter one yes so just just a question yeah go ahead yeah uh yeah sorry so sorry the cohort one I think that's what we discussed right so it's essentially uh trying to maintain some use races so whatever policies uh who have performed should eventually bring the system to that ideal homeostatic perspective right so uh why couldn't we just assume like a normal distribution so even if it is fat tailed for example if the empirical distribution is like the key uh distribution in this uh formula is you know has a heavy left tail you can essentially bring it back towards that by essentially modifying the policies I thought that that was the context of this right yes broadly you're right so when we look at the full free energy expected free energy formalization so here's equation 7.4 or equation 2.6 you're absolutely right that there's a pragmatic value that's based around reducing Divergence between observations and preferences but in equation the one we've been focusing on and trying to really unpack because again also like otilde is used in pragmatic value we're only talking here I mean they would still be yeah yeah but but that would make sense right because uh your your conditioning on your actions having some uh effect on the state on your own State and your state in the world or maybe some state in the world so you have to continuously monitor until it comes down to the expected observation that's what the generative model would spit out right so it makes perfect sense and it's completely coherent in that way right yes I hope and slash expect and prefer that to be true what you're describing about like bringing the homeostatic variable closer to preferences expectations that is pragmatic yeah but that would also yeah so that wouldn't it wouldn't be surprising that hotel does that and not be distribution as well as what I'm saying no you wouldn't need it to be there yeah you're right it is needed to be there because these are surprises and entropies and so on about observations um and then these this part is really digging into um decompositions of epistemic value so these are there might be two policies that are both expected to bring your temperature down or whatever but they have different epistemic values does it matter if it just brings your temperature down well it would look the same um essentially it would have the same effect so why does it matter uh why why uh did we have to have this kind of uh yeah so there will be multiple policies that

would bring it down bring the temperature down if you're if the preferences to bring the temperature down it does not matter which policy is chosen right so that's essentially what I think it's being said okay so yeah as long as you minimize the expected here's two investment opportunities from a hundred dollars 150 plus or minus one or 150 plus or minus 200. which one do you want that's not what you were discussing um so it is because they have space is very different no it's not the same thing they are the payoff space expected ambiguity in the 200 plus or minus 200 is very very different or and your Mutual information will be very different because the payoff space is different exactly you are essentially taking yeah but uh if something is going to bring down uh temperature plus or minus two plus or minus four I mean it's not a huge difference as long as it brings it down right so that that's essentially what I'm getting at so okay why does it matter okay so in this in this modeling the body's expected temperature is like the expected return on investment and the variance around the expected temperature is like the variance around the S uh Investments and so right but they wouldn't be the same policy is what I'm saying sorry they won't be the same payoff space in the second case so if something uh generates a hundred dollars but has uh a low probability of going bus so that there's a high variance that's not the same payoff as something that reduces the same hundred dollars but is that has a lower variance the payoff space spaces are different and that will be captured in that uh coolback Divergence that you have there that's right I would prefer the one with the lower that's right that's right because the payoffs yeah so uh but uh I think the discussion was around if multiple things lead to the same epistemic value and they have the same payoff why does it matter which one is taken rate is that last piece there might be multiple policies with the same and then then epistemic value is the Difference Maker that favors policies that have more clarity about observation right right so pragmatic value would say would would say that's not necessarily true right uh that's not necessarily true let's say that uh okay when the payoff space is completely unknown that the steps that you're taking uh at T equals one two three will bring down the uh we'll we'll have the same pragmatic value but one of these actions will lead to some very bad consequences that step peoples 15 because you're giving up something else because if we're talking about temperature we're in the space of temperature okay so yeah all right all right let's uh so let's just bring it back to the finance okay the space is dollars that's the dollar space uh yeah okay so we could use dollars numeric fine so we benchmarked the dollars how many dollars in making profit so I have two Investments uh one uh both of them generate a hundred dollars but one of them requires me to borrow a hundred dollars in order to generate a hundred dollars the other one will generate a hundred

dollars over say 50 time steps this would be quicker the second one where you borrow 100 so it would be quicker because you're borrowing a hundred dollars but there's an interest rate that you have to uh pay out so there's a much uh so you will have to generate much more than a hundred dollars in order to for this to be viable okay because you have an interest rate payout okay I'm gonna try to follow up here policies are defined in terms of sequences of affordances to take so the affordance here that you're highlighting is borrowing or not yes but there are actions that would be like if you have a very high fever in order to bring down that high fever sometimes it makes sense to borrow that hundred dollars whatever metaphorical variation of that borrowing that you have to bring it down to something more manageable right so it depends on how close you are to some sort of threshold where it's intolerable uh I mean putting it in in more mundane terms that hey I have I have a bad toothache it doesn't make sense for me to borrow a hundred dollars to fix my tooth I could just brush my teeth and hope that it goes away but I have a knife in my back I need to get to the hospital it makes sense to pay a hundred dollars because you're gonna die if you don't look okay let me try to try to see this is how I'm seeing that is when we're um near homeostasis then yeah we take actions that basically keep us there and we try not to like add risk to the situation whereas near the limits of our homeostatic tolerance we may engage in high risk behavior because there's like already a non-negligible like because of payoff space is much larger yeah the payoff space the positive payoff based on life so when you are when you're closer to when you're closer when you have a knife in your back the payoff from borrowing a hundred dollars will save your life but you have a toothache borrowing a hundred dollars might cause more risk that you go bankrupt right but you won't you will still survive if you have the tooth a toothache you're just short a hundred dollars which you could have used elsewhere there's no way you can use that hundred dollars if you're dead yeah so it does it does a much higher payoff borrowing at that point than it is with the toothache example rates it depends on where you take these actions I got you I'm seeing the space are the axes so here we have dollars and life and so you're saying that yeah but it is a multi-dimensional uh this one so if you think about it as let's say yeah so uh let's make it more concrete right so if you have a drone uh and uh you have a trade-off between battery life and staying in the air uh and let's say one of your propellers goes out uh you know just redistributing power to the other mortars so that you stay in the air even though it reduces your uh hover time makes more sense because you could Land versus you know uh trying to fly around and then crash eventually because you've lost one motor so but whereas let's say that uh it's so the other option would be let's just go faster to a certain destination then it

doesn't really make sense to do so you're just reducing your battery life to get to a destination faster but but if something happens on the way which causes you to lose your motor then you don't have enough battery in order to enough battery power to distribute to the automotive so that you can hover and come down safely uh does it make sense uh Eric so this you know I wonder if we could table this particular discussion it seems to be going um you know and get back to some of the other questions in the chapter I mean if I I actually we we had a question that was pending from last week that I think is kind of critical to the understanding of the chapter and I wonder if we could so you want to make sure we have time for it running out of time here yeah and that's the the question about what the relative role is of alert how it actually operates that you learn something that the rat learn some information prior to deciding which of the two branches of the maze to take um and the way they seem to frame the chapter is that well they set it up so that the rat um has its preference for epistemic Value because of epistemic value of learning which of the two branches to take um is a is a bigger term it'll first go to the bottom branch um learn which of the two top branches has the payoff then once that's resolved it knows that then it go then it decides which of the top branches to go to and it gets its reward so my understanding of that is that and this is my claim in this question is that um the rat is has a preference to resolve epistemic uncertainty regardless of whether it's useful or not so if you added more uncertain questions like you know um what color is um you know my Apple today but there's no there's no apple to be gotten it'll still want to resolve that question and any other questions because those are uncertainties so the epistemic value it'll collect that epistemic value and then finally it gets around to collecting its reward um that's in contrast to a model where the purpose of resolving epistemic uncertainty is to gain the reward which requires look ahead so in other words the the planner or the policy would be I try to learn what's needed to collect the reward I do that and that's posted as as epistemic value with a purpose then the the order of operation is okay I figure out what I need to know I learn what I need to know then I go and get collect do the exploitation so I don't that's what I would expect that this kind of a framing would give us but I don't see that that's what you're actually giving us all right here here's how I see that the um example that we would want to see would have a bunch of um uncertainty resolving this one picks a number between 1 and 10 and tells you what it is this one's a number between one and a million so it's an it's an incredible information resolving button then the question is can the relevant sources of information be sought after I think uh the examples is prepared in several ways for example the rat already knows the semantics that this is related to here that's been just the structure of the model is

already preparing that any any any uncertainty reduction about this like basically it's already encoded in the model that this bottom cue is information about this so this is kind of like saying um and this relates earlier like not only are the those those like implicit structurally encoded aspects of knowledge but also there's parameterization questions like we discussed if it really pers if it has a dire urgency for food to um the earlier questions about like the urgency of the imperative it may just go for it and say I'd rather have a 50 50 shot now than a 98 in two time steps so it depends on how it's parameterized but even more deeply it depends on the construction of the model what kinds of relationships are implicitly and explicitly being linked and like that's a a it's like a deep level of modeling to understand what would be like a more neutral way to frame this question and it's like well how many layers back do you want to pull like you could have a pre-play where a rat has um an association Matrix there's three information sources and it's allowed to freely explore without any shock or food and determine which one of these information sources has like a causal relationship to these edges and then there's a learned cue that now we're in game time and now it's going to rely on its past learning about which information resource is relevant for the pragmatic and then it's I mean there's just how many layers back does one need to pull before the rabbit is not in the Hat I think is going to be a serious because even toy demonstrations have been seen as slam dunk um they're not well instead of bringing up Palm DPS here Palm DPS are really good for planting but they don't have this rat do any planning it seems like they're really not aspiring very strongly to build a smart rat here and they're not achieving it either yeah it's um true because the first movement again presuming that the parameterization is such that epistemic value is salient the first move is dominated by epistemic value and they say well now that that value has been tapped now it can step into pursuing pragmatic value with even increased confidence but it's not considering this the set of time Horizon 2 policies that is not being explicitly encoded or that means that even implicit yeah these are these are transition matrices these are just the four options this is this is depending on where you are you can if you're you know you can stay you can go down yeah does section seven three address planning and if it doesn't why do we need G why can't we just use a one-step variational free energy approximation what's it's it's saying what's the most likely thing for me for me to do right now is a question about now in the past well I think the answer to the question is exactly as you said it before which is they baked in this particular hardwired rat for which in this case it it the best thing to do is to First explore learn and then exploit so they designed it to do only this very by wrote to do the right thing as opposed to the rat actually use it doing any look ahead or

any intelligence to do the right thing yeah it's structurally implicitly hardwired and then it's fine-tuned because again there's structures where it wouldn't do this and there's tunings where it wouldn't do this but there's there's um it's cut kind of in in the tens place and in the decimal point this example exists in a very limited manifold of models and parameterizations where one step optimal policies emulate two-step planning like I just happened to love sacrificing pawns and taking castles so I'm willing to sacrifice upon so that later I can do a castle but of course you can't walk around with the belief that you love sacrificing Pawns I actually wondered why there was an enumeration of policies in the planning section learning hyper priors hidden States on hidden States seeing hidden States as outcomes of other hidden States so that they can be learned or fixed a little bit of technical details on um on a multi-parameter minimization Theta being just a vector of parameters for the generative model and dear Schley distributions which I think could be gotten into but that's kind of distribution specific creatures select the most appropriate data to improve their generative models I think there's more that we can dig into and explore on similar wavelengths like risk plus ambiguity but then now there's an Information Gain I thought we just had risk plus ambiguity then we didn't really yeah people point to structure learning but don't they always as a way to get around these questions but then it's the structure of the structure learner and then people end up with these Turing machines that don't plan people hope that Bayesian model reduction will be attractable and provide heuristics for structure learning but I haven't seen any empirical examples that come to mind where Beijing model reduction was used to identify actionable lower dimensional useful models but we know that structure learning on the state space of hierarchical models is going to be essential active inference or Beyond because people thought that the explosions the the computational complexity class of just branching time active inference or anything within a model this is going to be like exploding upon that by several exponents and we have these fundamental questions about the continuous time interpolations why are there continuous interpolations in chapter seven but it's the discrete Time chapter next time we come to chapter eight I think this will be quite interesting we're going to talk about dynamical systems motor control um lockabletera we'll have some Justified continuous lines um LaPlace assumption Lorenz live stream number 32 stochastic chaos and Markov blankets Hybrid models with discrete continuous Fusion some advances in continuous time thanks everybody see you soon thanks everyone